

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

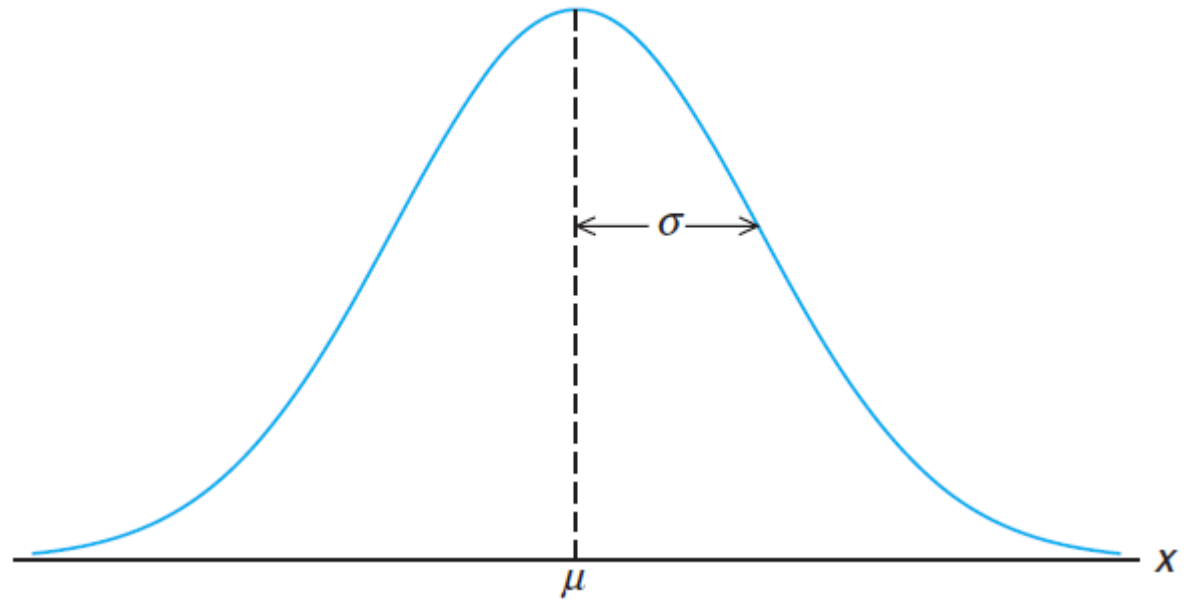
وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Continuous Random Variable (Normal Distribution)

Week 11

Normal Distribution

- ❖ Abraham DeMoivre (1733)
- ❖ Gaussian Distribution
- ❖ Bell-shaped curve
- ❖ Real phenomenon (Nature, Industry, Research)
- ❖ Two parameters
- ❖ Basis for statistical Inference



$$X \sim N(\mu, \sigma^2)$$

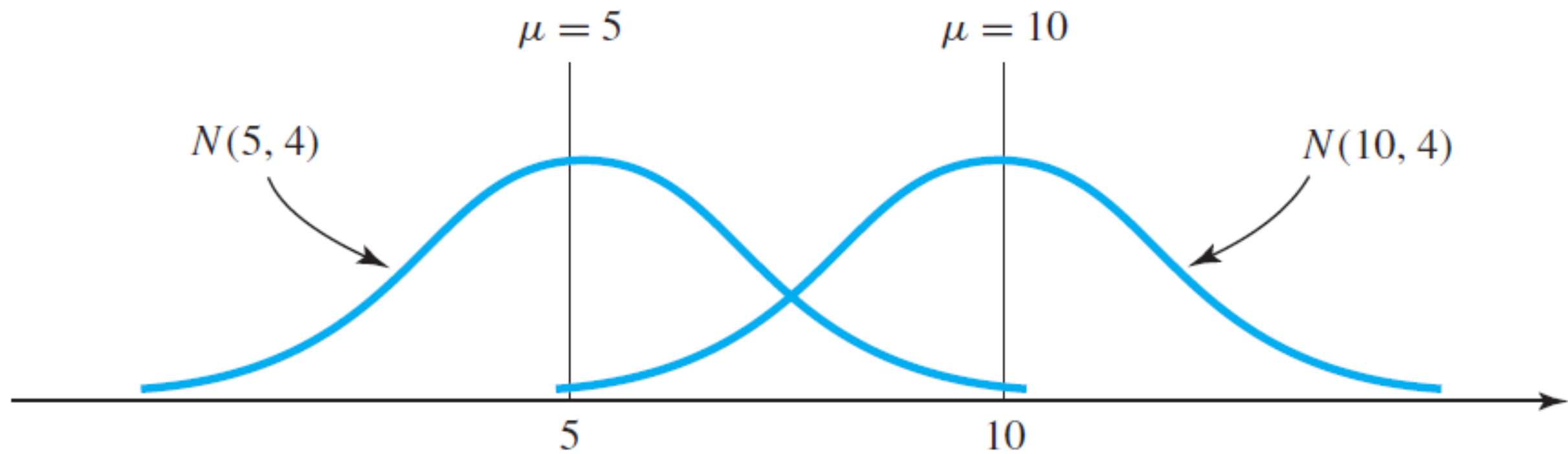
The Normal Distribution

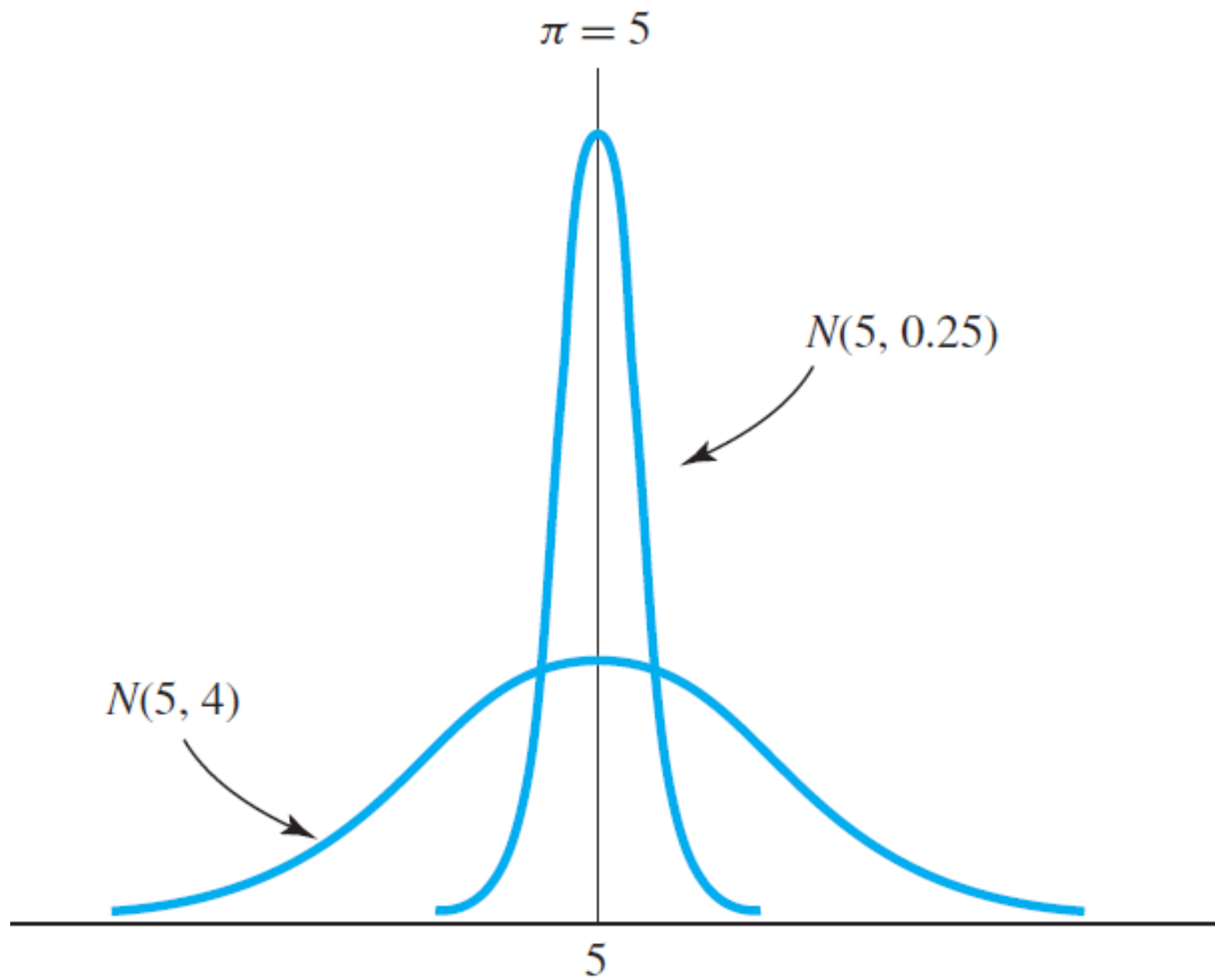
The **normal** or **Gaussian distribution** has a probability density function

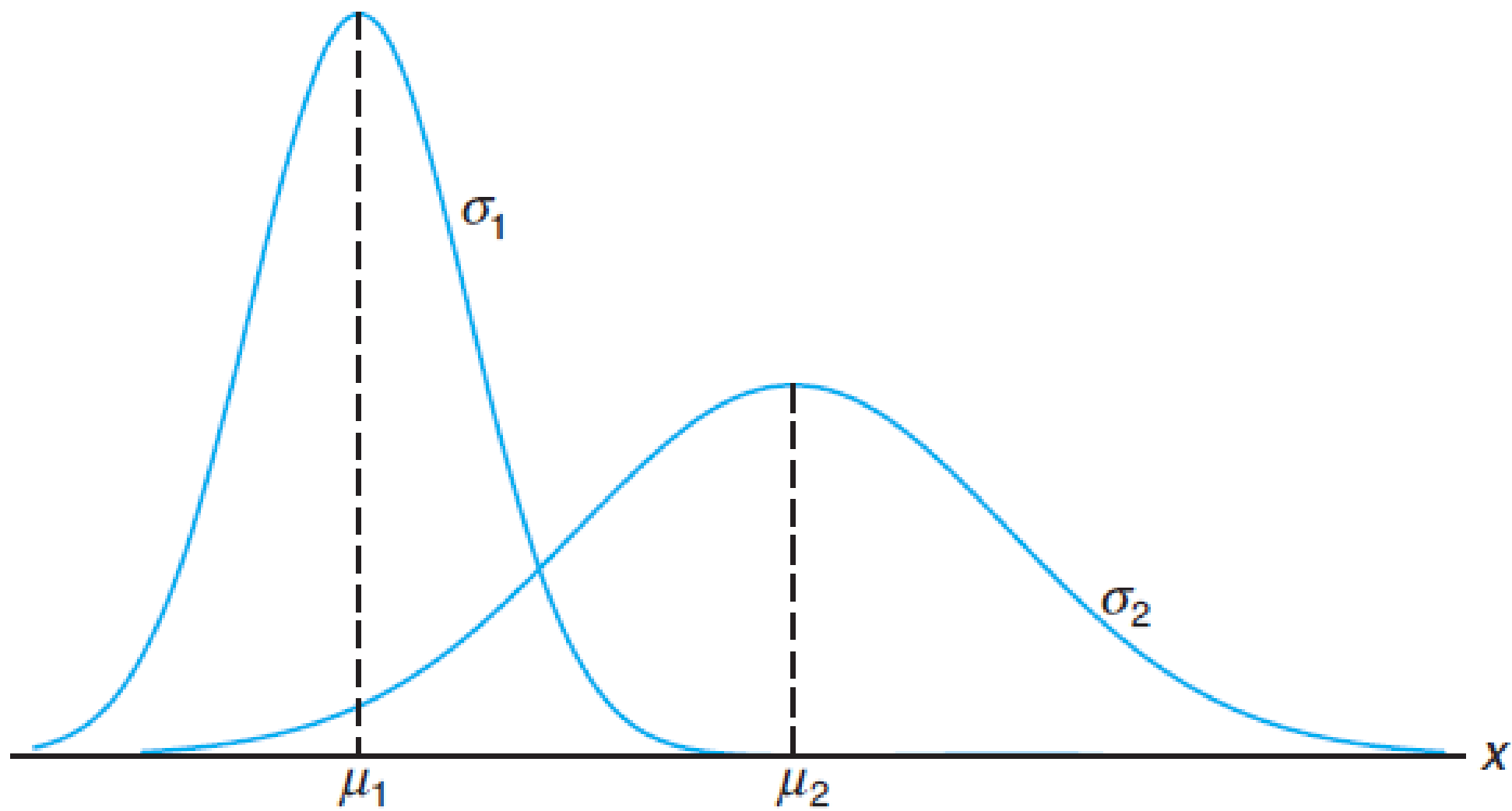
$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

for $-\infty \leq x \leq \infty$, depending upon two parameters, the mean and the variance

$$E(X) = \mu \quad \text{and} \quad \text{Var}(X) = \sigma^2$$





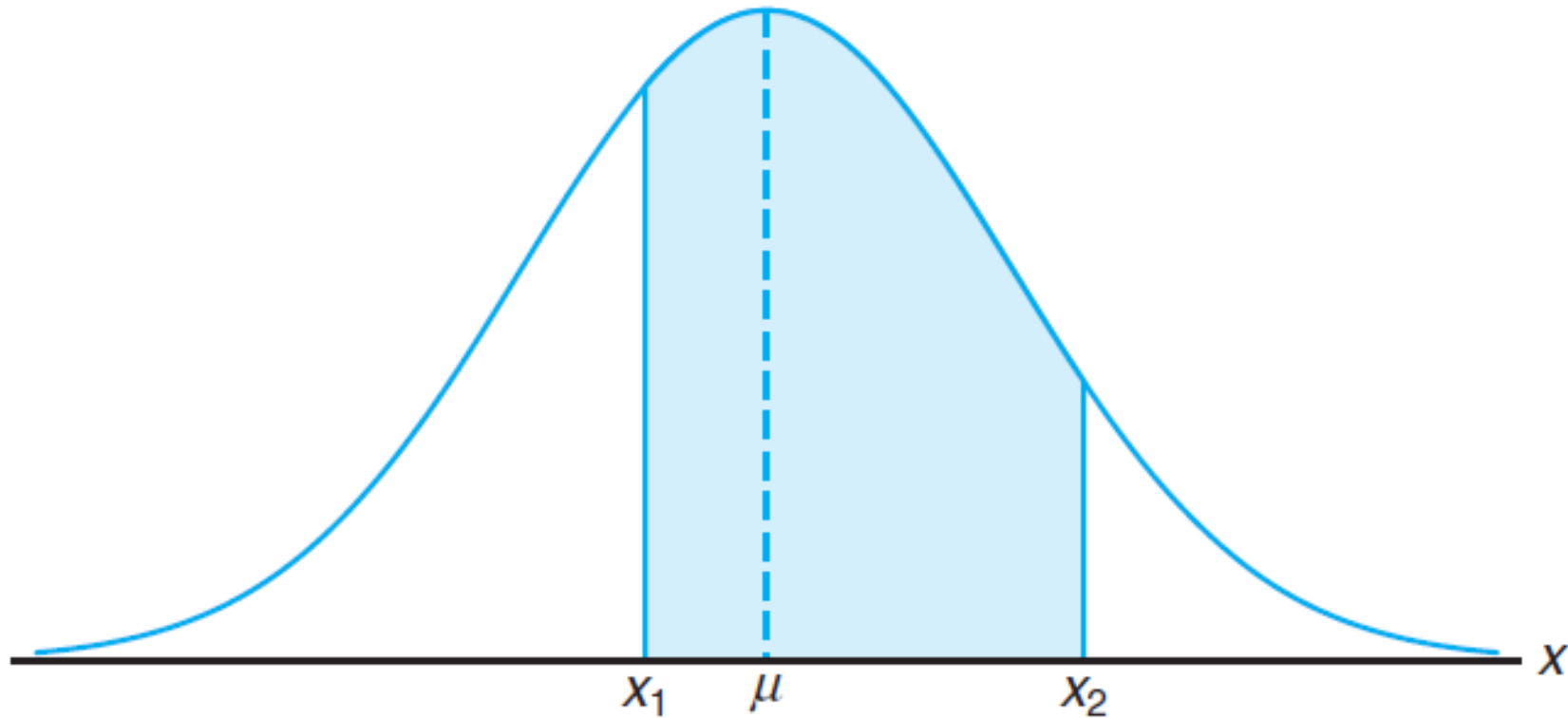


Properties of Normal Distribution

1. The mode, which is the point on the horizontal axis where the curve is a maximum, occurs at $x = \mu$.
2. The curve is symmetric about a vertical axis through the mean μ .
3. The curve has its points of inflection at $x = \mu \pm \sigma$; it is concave downward if $\mu - \sigma < X < \mu + \sigma$ and is concave upward otherwise.
4. The normal curve approaches the horizontal axis asymptotically as we proceed in either direction away from the mean.
5. The total area under the curve and above the horizontal axis is equal to 1.

Area under the normal curve

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} n(x; \mu, \sigma) dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{x_1}^{x_2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$



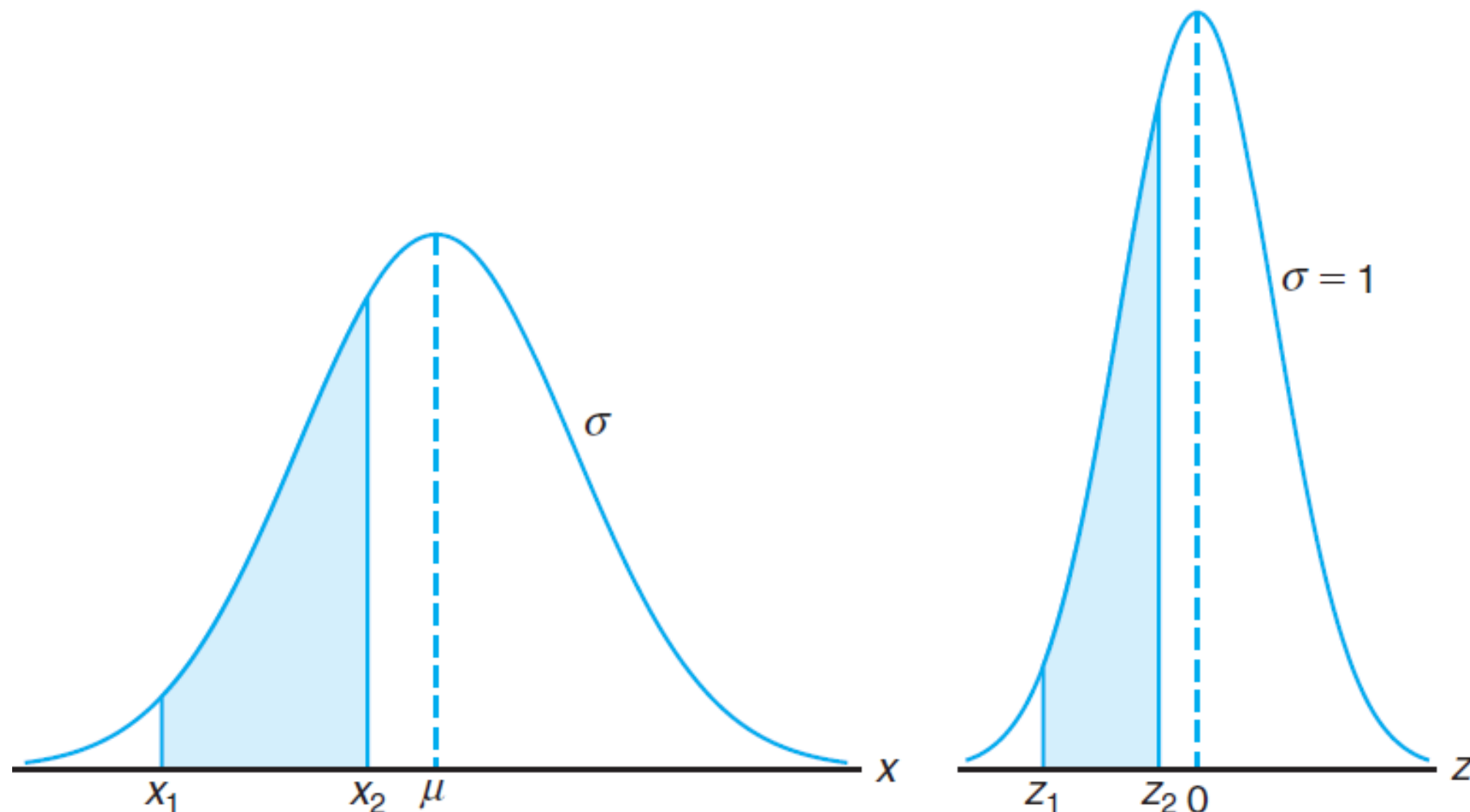
Relation between X and Z

$$P(x_1 \leq X \leq x_2) = P\left(\frac{x_1 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{x_2 - \mu}{\sigma}\right) = P(z_1 \leq Z \leq z_2)$$

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \sim N(0, 1)$$

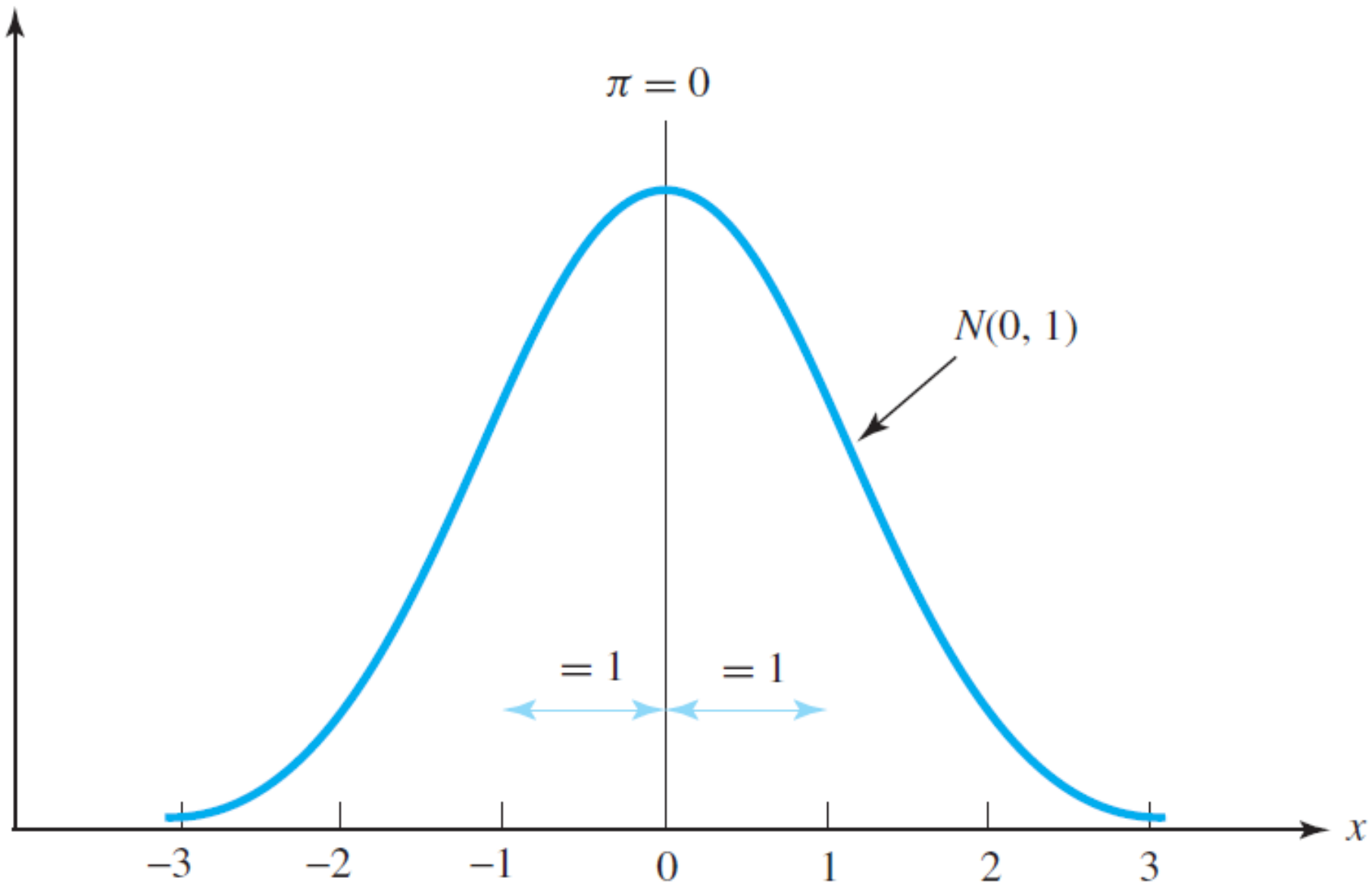


Standard Normal Distribution

A normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$ is known as the **standard normal distribution**. Its probability density function has the notation $\phi(x)$ and is given by

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \text{for } -\infty \leq x \leq \infty.$$

$$f(x) = \frac{1}{\sqrt{2}} e^{-x^2/2}$$

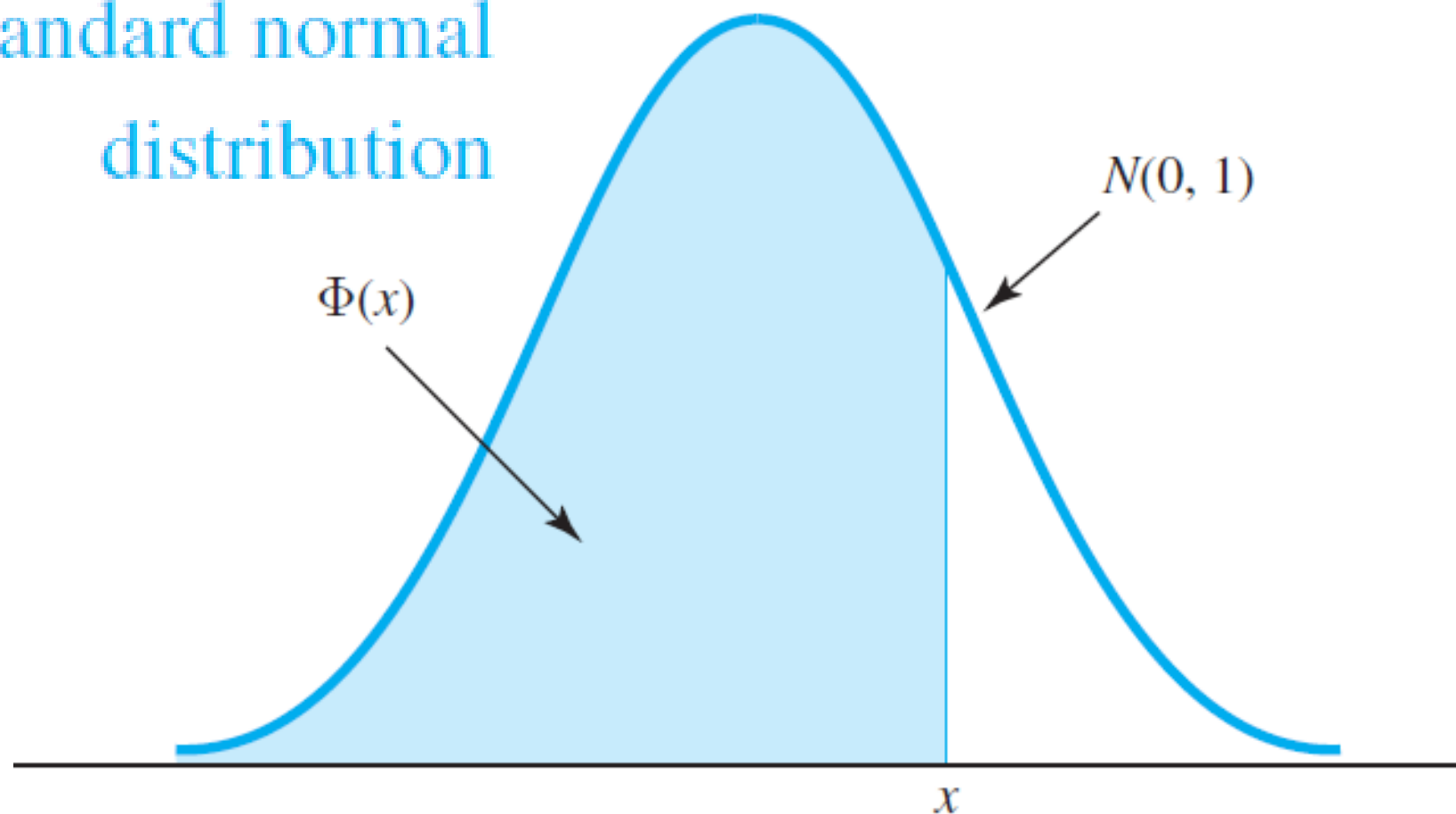


$\Phi(x)$ is the cumulative distribution function of a standard normal distribution

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \sim N(0, 1)$$



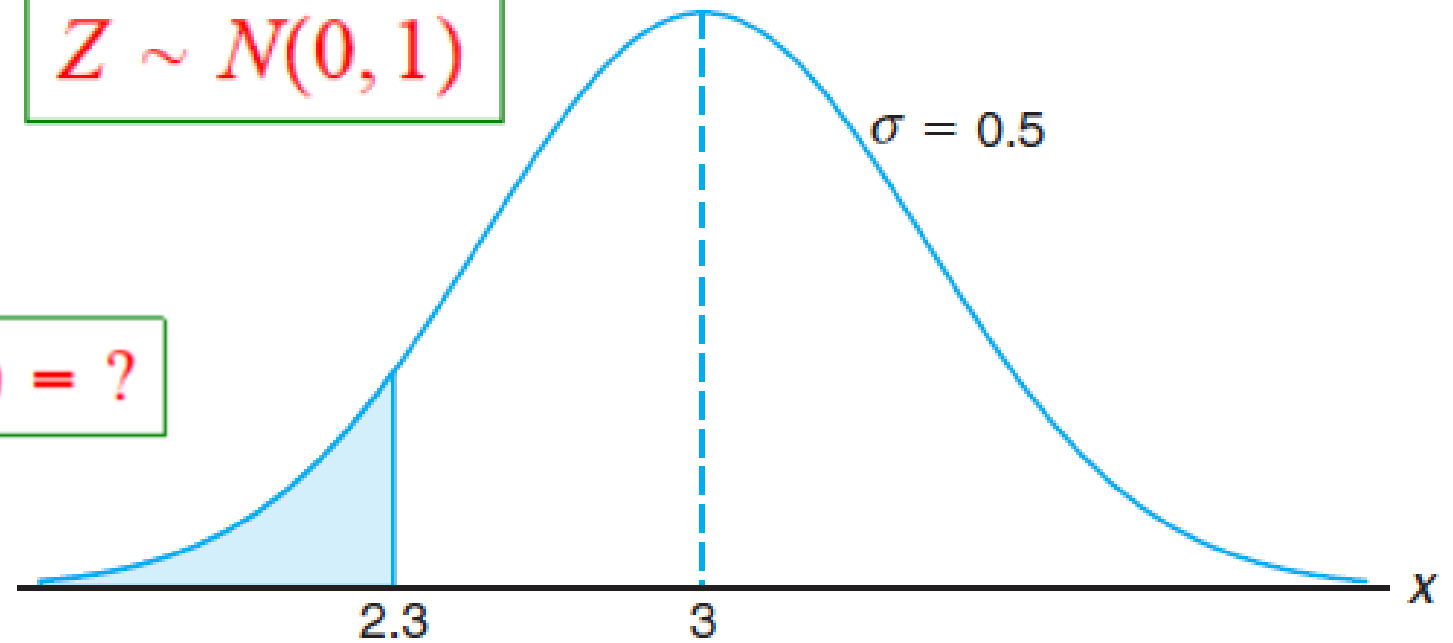
$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \sim N(0, 1)$$

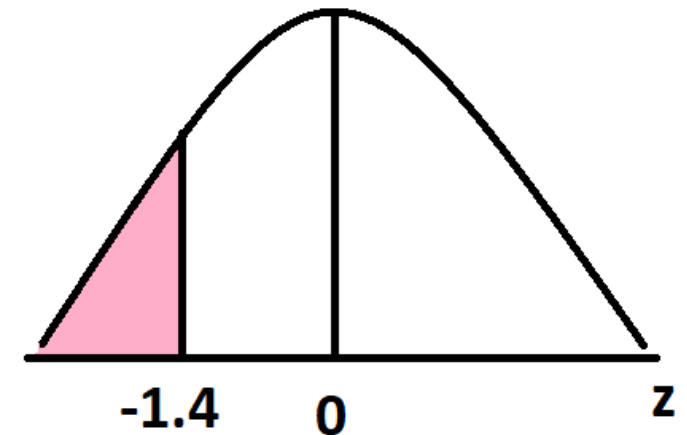
$$X \sim N(3, 0.5^2)$$

$$P(X < 2.3) = ?$$



$$P(X < 2.3) = P\left(\frac{X - \mu}{\sigma} < \frac{2.3 - 3}{0.5}\right) = P(Z < -1.4)$$

$$P(Z < -1.40) = 0.0808$$



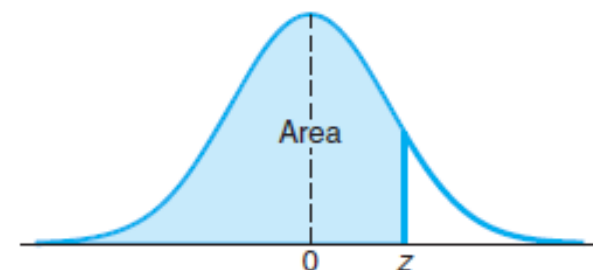


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

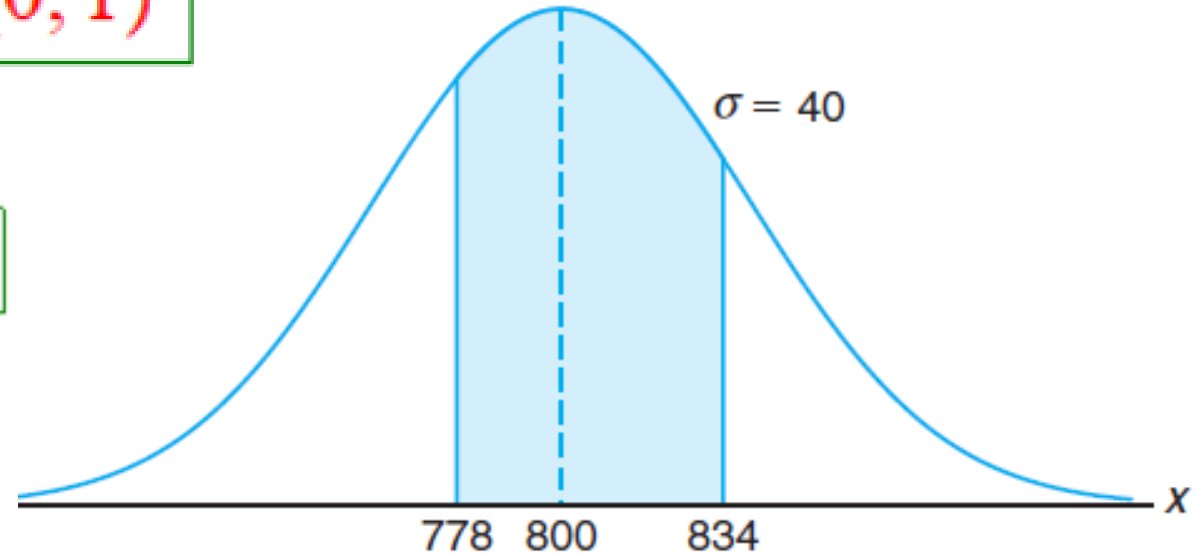
$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z \sim N(0, 1)$$

$$X \sim N(800, 40^2)$$

$$P(778 < X < 834) = ?$$

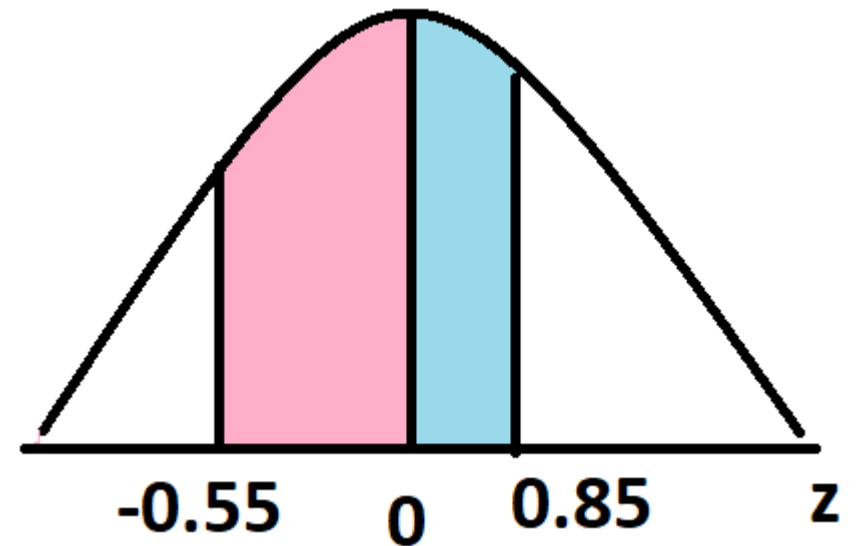


$$P(778 < X < 834) = P(-0.55 < Z < 0.85)$$

$$= \Phi(0.85) - \Phi(-0.55)$$

$$= 0.8023 - 0.2912$$

$$= 0.5111$$



$$\Phi(0.85) - \Phi(-0.55)$$

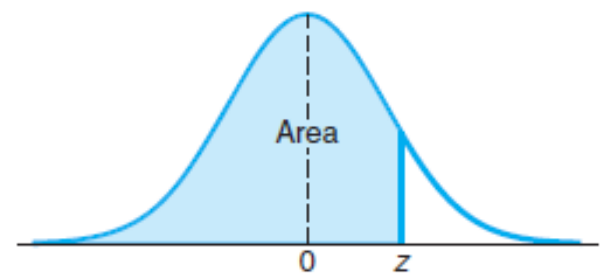


Table A.3 Areas under the Normal Curve

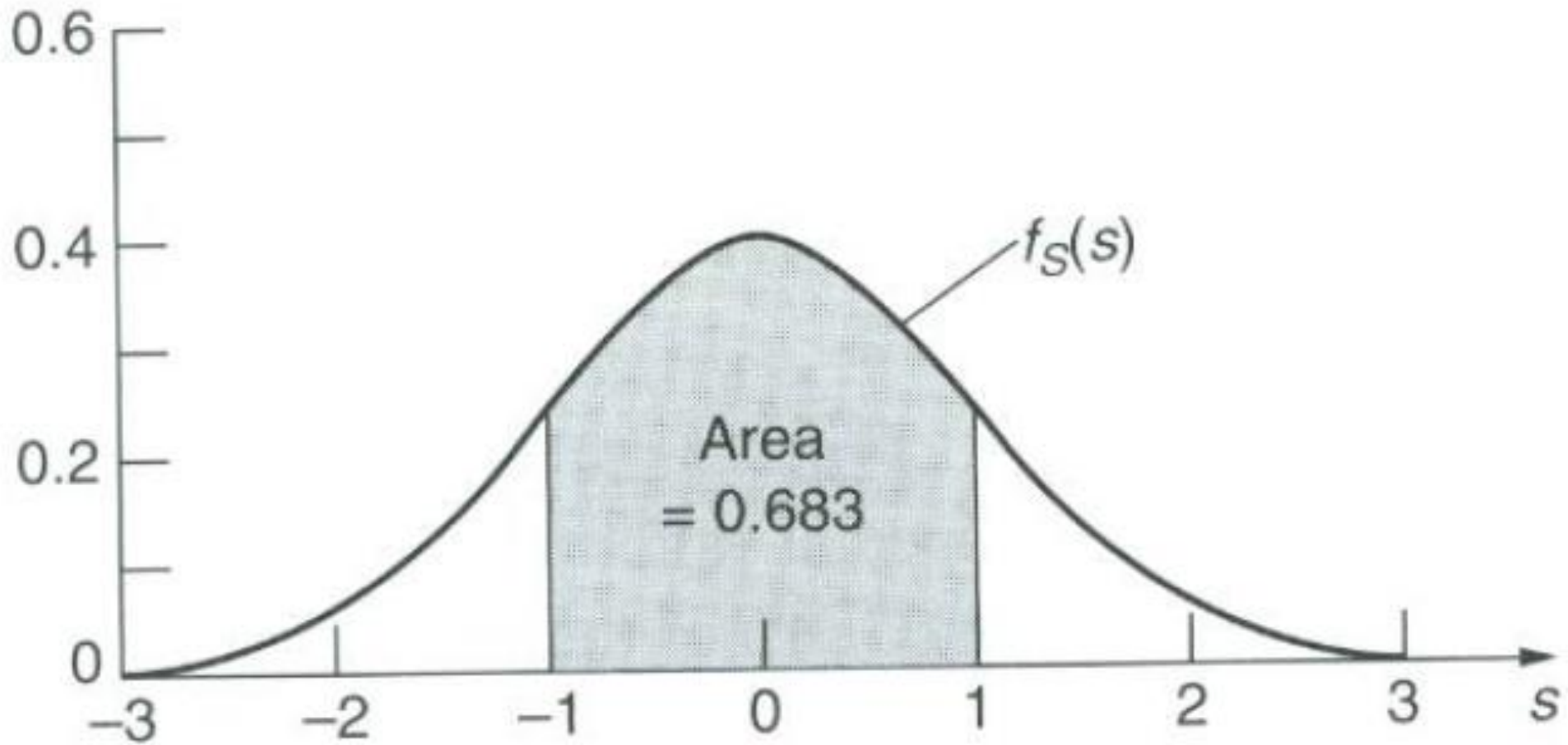
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776

Table A.3 (continued) Areas under the Normal Curve

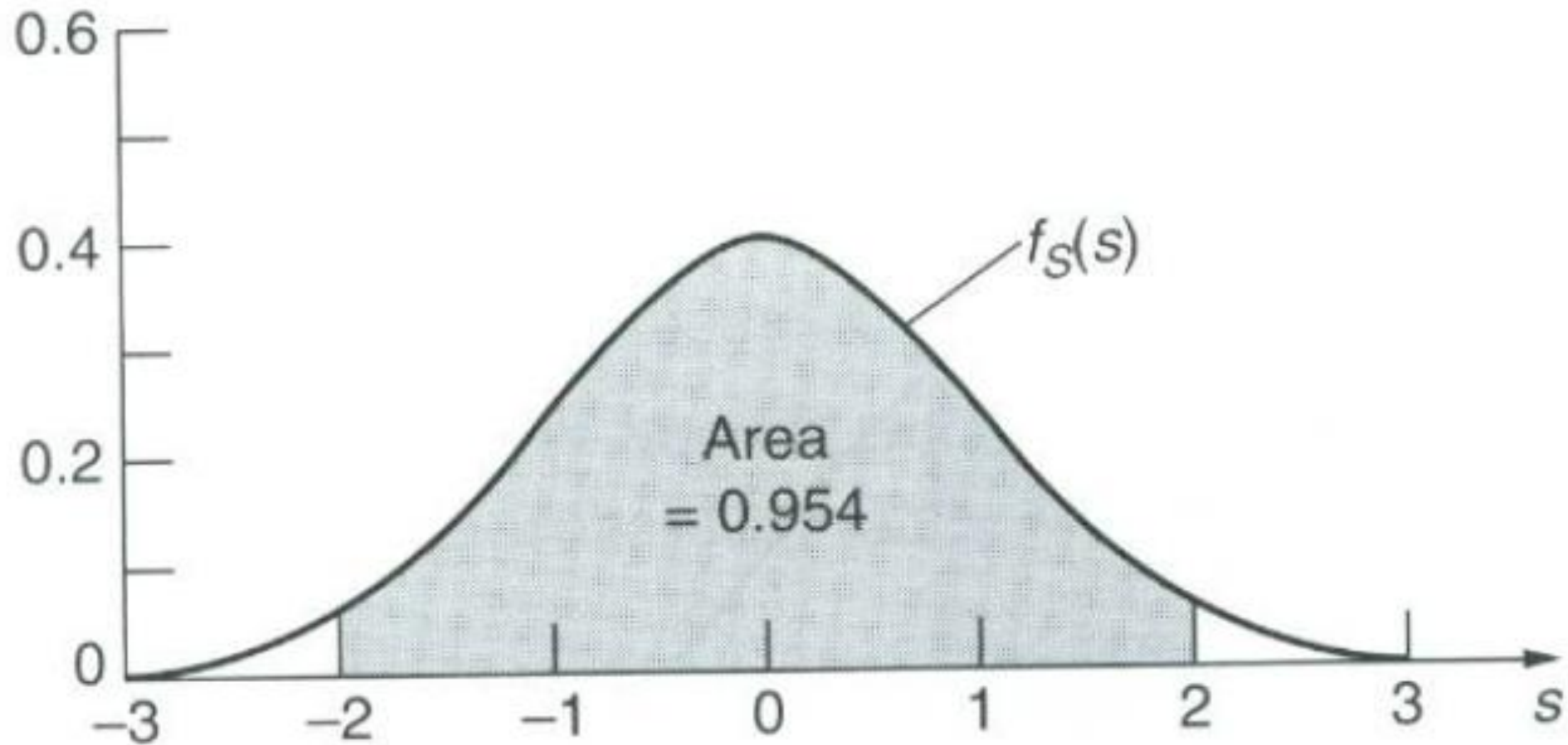
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389

$$\Phi(0.85) - \Phi(-0.55)$$

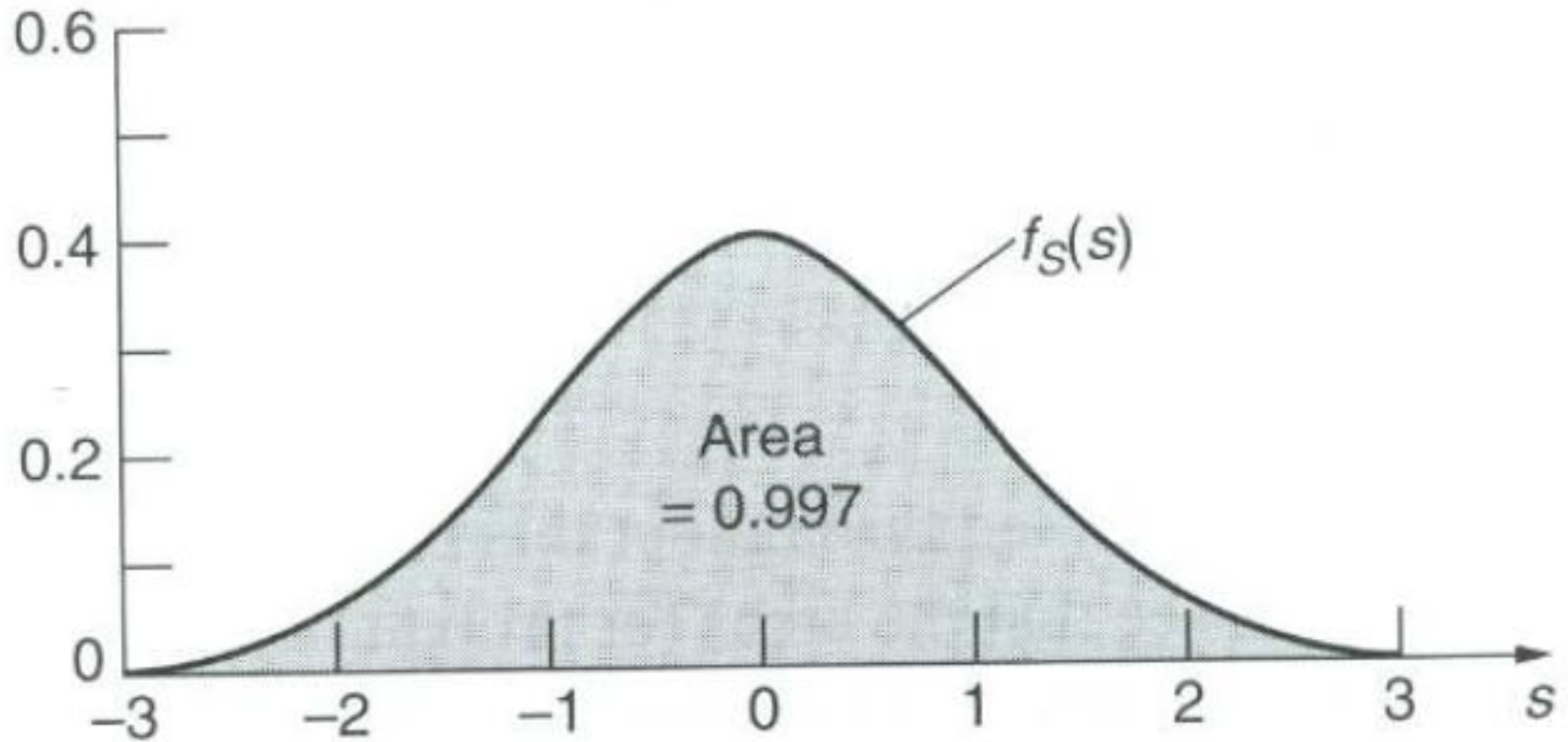
PDF of $N(0,1)$ covering 1 Sigma



PDF of $N(0,1)$ covering 2 Sigma



PDF of $N(0,1)$ covering 3 Sigma



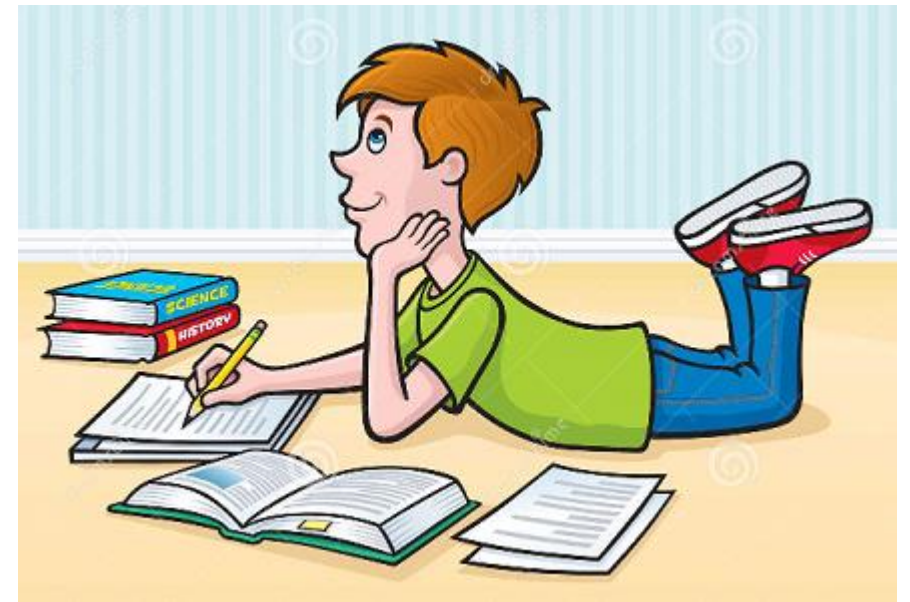
Task

$$X \sim N(3, 0.005^2)$$

$$P(X < 2.99) = ?$$

$$P(X > 3.01) = ?$$

$$P(2.99 < X < 3.01) = ?$$



$$X \sim N(1.5, 0.2^2)$$

$$P(X < 1.108) = ?$$

$$P(X > 1.892) = ?$$

$$P(1.304 < X < 3.01) = ?$$

Remember

When to apply Poisson Approximation to Binomial?



n is large and p close to 0, 1

Normal Approximation to Binomial

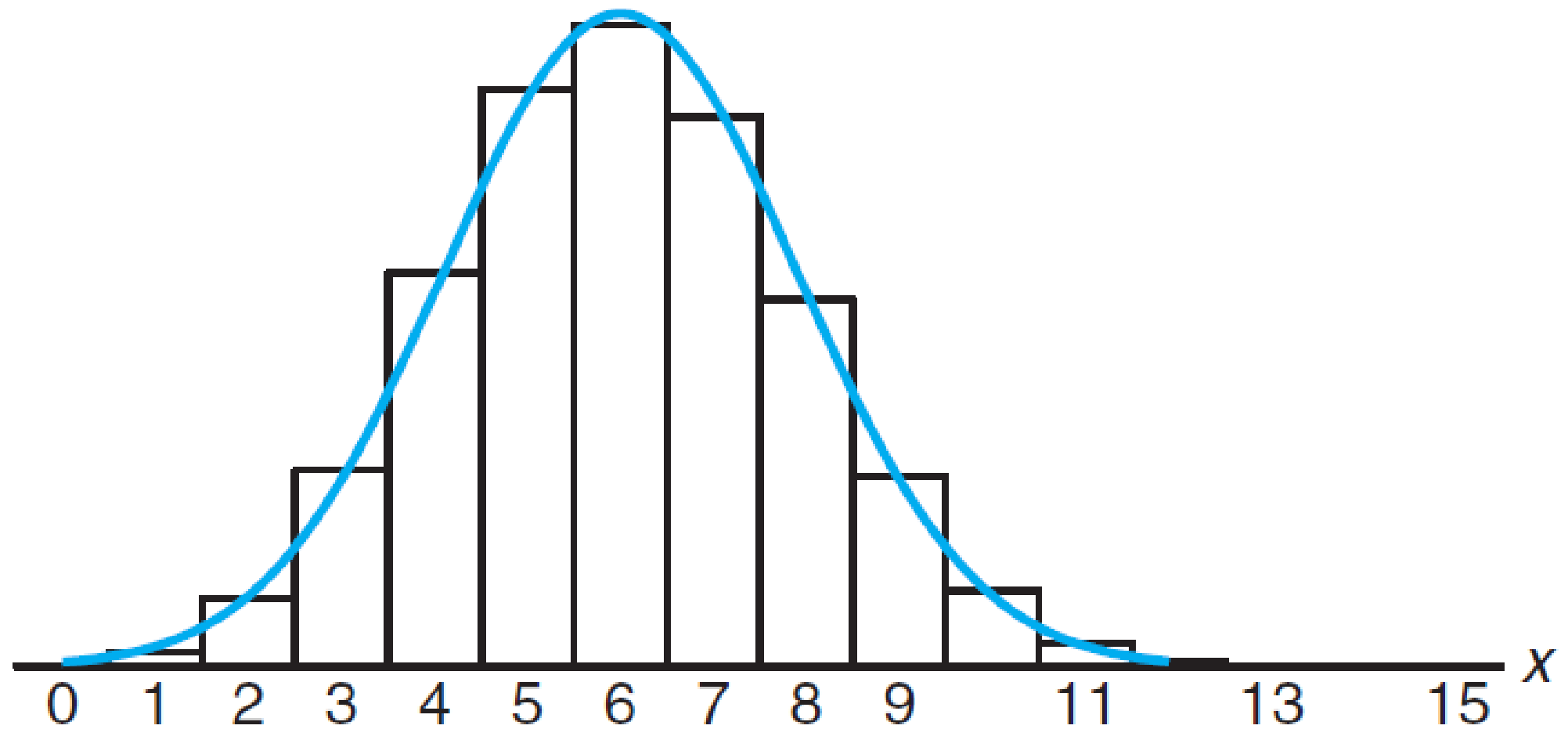
Either n is large or small with p close to 0.5

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the distribution of

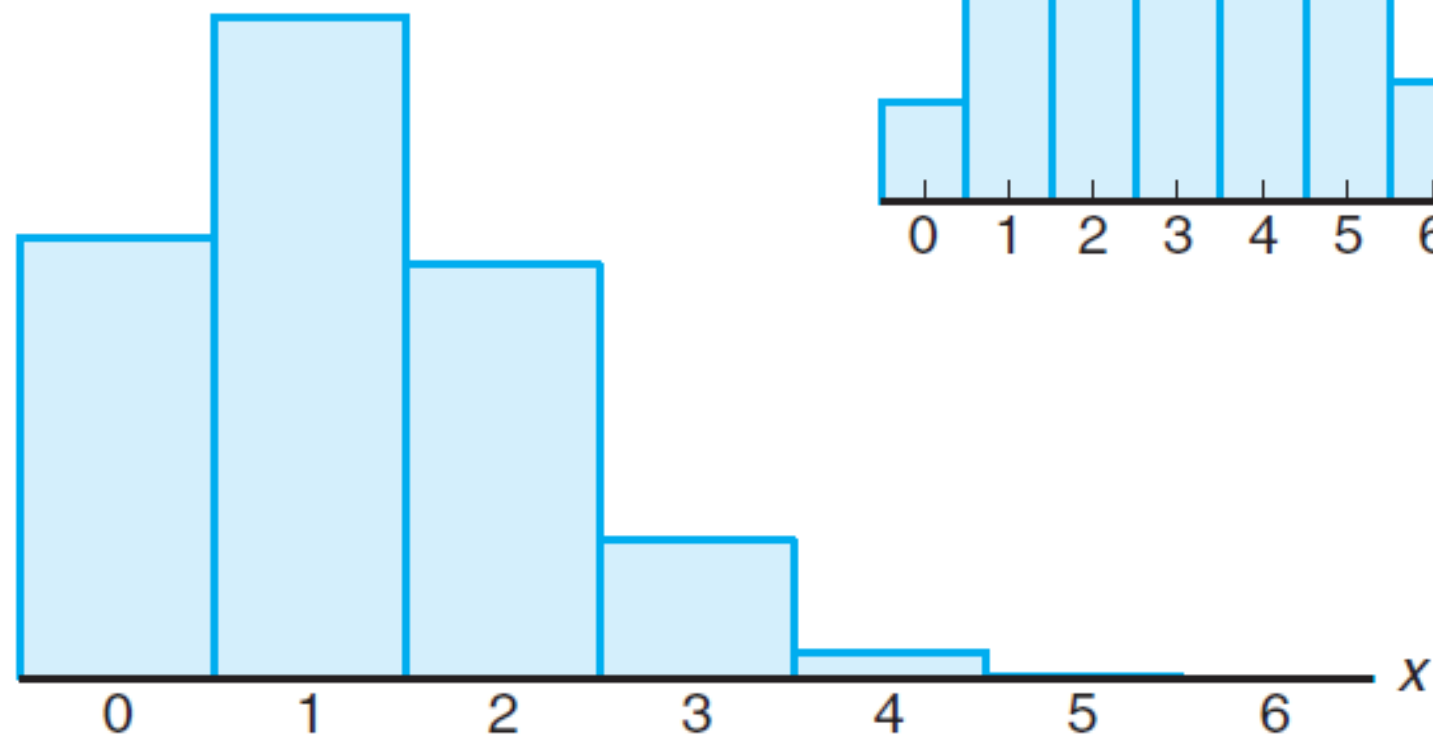
$$Z = \frac{X - np}{\sqrt{npq}},$$

as $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

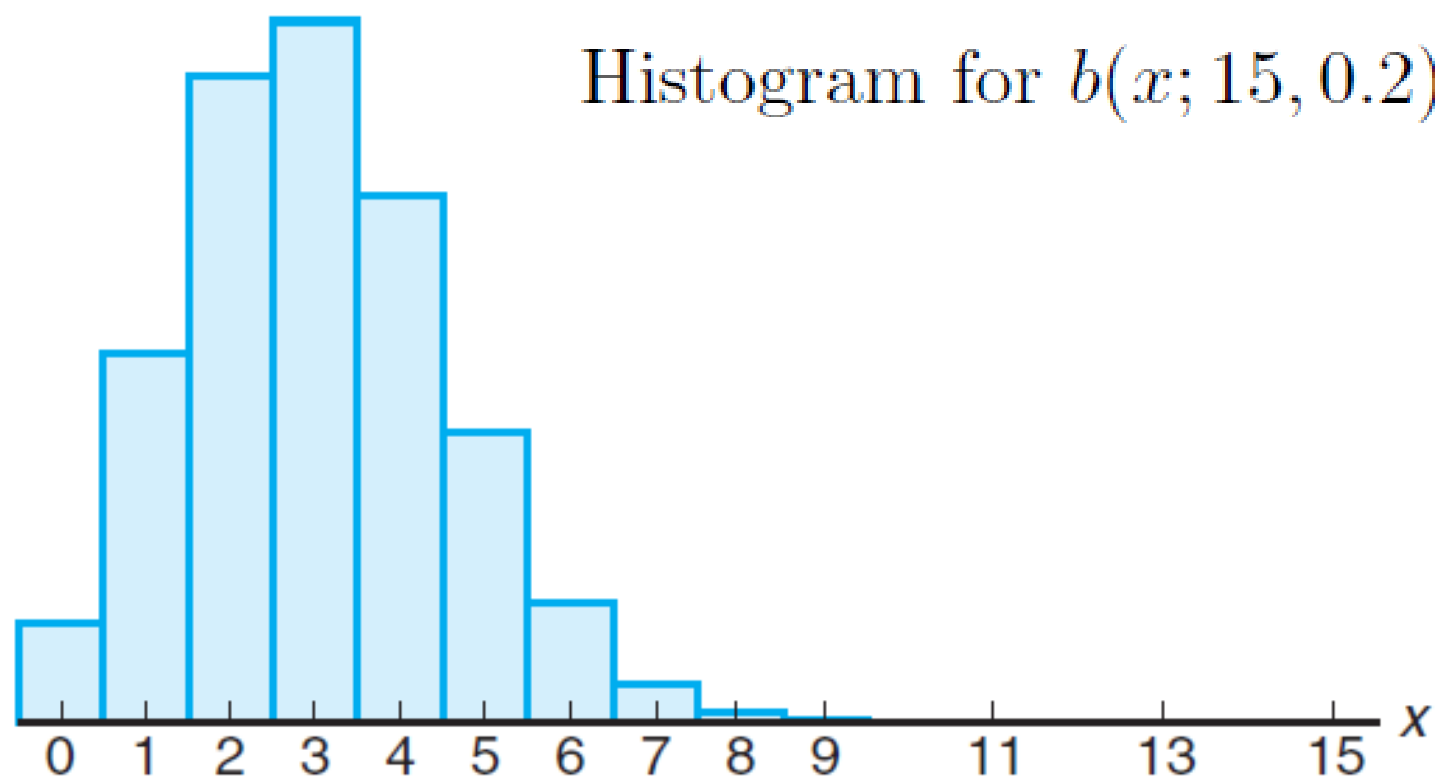
Normal approximation of $b(x; 15, 0.4)$.



Histogram for $b(x; 6, 0.2)$.



Histogram for $b(x; 15, 0.2)$



Example: 6

- A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?
- The probability of guessing a correct answer for each of the 80 questions is $p = 1/4$. If X represents the number of correct answers resulting from guesswork, then

$$P(25 \leq X \leq 30) = P(25 \leq X \leq 30) = \sum_{x=25}^{30} b(x; 80, 1/4).$$

Using the normal curve approximation with

$$\mu = np = (80) \left(\frac{1}{4} \right) = 20$$

and

$$\sigma = \sqrt{npq} = \sqrt{(80)(1/4)(3/4)} = 3.873,$$

Example: 6

A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?

$$\mu = np = (80) \left(\frac{1}{4} \right) = 20 \qquad \sigma = \sqrt{npq} = \sqrt{(80)(1/4)(3/4)} = 3.873$$

$$P(25 \leq X \leq 30) =$$

$$z_1 = \frac{24.5 - 20}{3.873} = 1.16 \text{ and } z_2 = \frac{30.5 - 20}{3.873} = 2.71.$$

$$\begin{aligned} P(25 \leq X \leq 30) &= \sum_{x=25}^{30} b(x; 80, 0.25) \approx P(1.16 < Z < 2.71) \\ &= P(Z < 2.71) - P(Z < 1.16) = 0.9966 - 0.8770 = 0.1196. \end{aligned}$$

Linear Combination of Normal Random Variables

Linear Functions of a Normal Random Variable

If $X \sim N(\mu, \sigma^2)$ and a and b are constants, then

$$Y = aX + b \sim N(a\mu + b, a^2\sigma^2)$$

Two Normal Variates

The Sum of Two Independent Normal Random Variables

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent random variables, then

$$Y = X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Example: 7

The radius of a piston head (X_1) has a mean value of 30.00 mm and a standard deviation of 0.05 mm, and that inside radius of a cylinder (X_2) has a mean value of 30.25 mm and a standard deviation of 0.06 mm. Suppose that the piston head radius and the cylinder radius are taken to be normally distributed.

1. What can be the distribution for the gap between piston head and cylinder?

Let the gap between the piston head and the cylinder is $Y = X_2 - X_1$

$$\mu = 30.25 - 30.00 = 0.25$$

and a variance of

$$\sigma^2 = 0.05^2 + 0.06^2 = 0.0061$$

$$Y \sim N(0.25, 0.0061)$$

Example: 7

The radius of a piston head (X_1) has a mean value of 30.00 mm and a standard deviation of 0.05 mm, and that inside radius of a cylinder (X_2) has a mean value of 30.25 mm and a standard deviation of 0.06 mm. Suppose that the piston head radius and the cylinder radius are taken to be normally distributed.

2. Find the probability that piston head will not fit within the cylinder.

The probability that a piston head will not fit within a cylinder can then be calculated to be

$$P(Y \leq 0) = \Phi\left(\frac{0 - 0.25}{\sqrt{0.0061}}\right) = \Phi(-3.20) = 0.0007$$

Example: 7

The radius of a piston head (X_1) has a mean value of 30.00 mm and a standard deviation of 0.05 mm, and that inside radius of a cylinder (X_2) has a mean value of 30.25 mm and a standard deviation of 0.06 mm. Suppose that the piston head radius and the cylinder radius are taken to be normally distributed.

3. Suppose that a piston performs optimally when the gap is between 0.10 mm and 0.35 mm. Find the probability for optimum performance of piston.

Suppose that a piston performs optimally when the gap Y is between 0.10 mm and 0.35 mm. The probability that a piston performs optimally is then

$$\begin{aligned} P(0.10 \leq Y \leq 0.35) &= \Phi\left(\frac{0.35 - 0.25}{\sqrt{0.0061}}\right) - \Phi\left(\frac{0.10 - 0.25}{\sqrt{0.0061}}\right) \\ &= \Phi(1.28) - \Phi(-1.92) = 0.8997 - 0.0274 = 0.8723 \end{aligned}$$

“n” Normal Variates

Linear Combinations of Independent Normal Random Variables

If $X_i \sim N(\mu_i, \sigma_i^2)$, $1 \leq i \leq n$, are independent random variables and if a_i , $1 \leq i \leq n$, and b are constants, then

$$Y = a_1 X_1 + \cdots + a_n X_n + b \sim N(\mu, \sigma^2)$$

where

$$\mu = a_1 \mu_1 + \cdots + a_n \mu_n + b$$

and

$$\sigma^2 = a_1^2 \sigma_1^2 + \cdots + a_n^2 \sigma_n^2$$

Average of Normal Variates

Averaging Independent Normal Random Variables

If $X_i \sim N(\mu, \sigma^2)$, $1 \leq i \leq n$, are independent random variables, then their average $\bar{X}$ is distributed

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

The Central Limit Theorem

If $X_1, \dots, X_n$ is a sequence of independent identically distributed random variables with a mean μ and a variance σ^2 , then the distribution of their average $\bar{X}$ can be approximated by a

$$N\left(\mu, \frac{\sigma^2}{n}\right)$$

distribution. Similarly, the distribution of the sum $X_1 + \dots + X_n$ can be approximated by a

$$N(n\mu, n\sigma^2)$$

distribution.

TASK

The weights of a large number of miniature poodles are approximately normally distributed with a mean of 8 kilograms and a standard deviation of 0.9 kilogram. If measurements are recorded to the nearest tenth of a kilogram, find the fraction of these poodles with weights

- a. over 9.5 kilograms;
- b. of at most 8.6 kilograms;
- c. between 7.3 and 9.1 kilograms inclusive.
- d. P_{73}
- e. D_4

Task

Suppose that orders at a restaurant are iid (independent and identically distributed) random variables with mean \$8 and standard deviation \$2.

- Estimate the probability that the first 100 customers spend a total of more than \$840.
- Estimate the probability that the first 100 customers spend a total of between \$780 and \$820.
- After how many orders can we be 90% sure that the total spent by all the customers is more than \$1000.